

Part III

Low Back Rehabilitation

This part of the book presents better rehabilitation practices based on the evidence discussed in previous chapters. The chapters in part III discuss evaluating patients and developing beginner and advanced exercise and rehabilitation programs.

Chapter 8

Building Better Rehabilitation Programs for Low Back Injuries

All body systems depend on movement for health. For spine health, however, the traditional approaches for enhancing fitness must be reprioritized. For those with painful backs, attaining movement skill and competency must replace energy systems training that emphasizes cardiovascular and muscle components. Too many patients are taught that 45 minutes on the treadmill or stationary bike will rid them of pain. Also important is the notion that we are not born physical equals. A St. Bernard has little chance at the greyhound track except to become injured. But the greyhound attempting to carry a St. Bernard's load up a mountain will crumble. Both can enhance their pain-free athleticism, but they need different approaches.

Many people develop painful backs because of movement flaws. Lifting even extremely heavy loads can be accomplished safely by athletes with perfect form. But movement flaws cause repeated or prolonged loading that is abnormal for the tissue, so it slowly becomes painful. Part of the stabilization approach is to correct the aberrant patterns to metaphorically stop picking the scab. As a result, the tissues become less sensitized, the repertoire of pain-free tasks increases, and motion returns. This is why it is essential to perform therapeutic exercise pain free. The presence of pain also leads to substitution patterns as the spine literally learns to limp. These patterns must be corrected, and they require pain-free motion. So, don't worry about the concept of restoring function too soon. This retards progress. Address the painful tissues and then work on function. Choosing the correct movement tool is also necessary for back rehabilitation success.

Consider the action of rising from a chair. Clearly, a squat pattern is best. Now consider someone who must lie on the floor to perform an exercise. Should she choose a squat pattern to move to the floor? If she does, she will create large knee and low back forces, which, over time, will lead to the development of painful disorders. Instead, this person should choose a lunge, which spares the joints. For a person who has lost some of the ability to squat

deeply as a result of a damaged knee or back, or advanced age, the lunge pattern makes moving to the floor possible, and pain free.

Movement competency is different for different people. For example, a construction worker needs different movements, strength exertions, postures, and endurance than a mail carrier or a bank teller. A high school student wanting to enhance her volleyball spike desires more explosive hip power and core stability, but her grandfather needs to maintain his ability to step quickly after stubbing his toe to prevent falling and breaking a hip. Both want to enhance their movement skill. However, the student's training program to maximize her movement ability would be very different from that of her grandfather.

The definition of good movement changes as we grow, mature, and age. Moving well reduces joint load, in turn reducing the risk of degenerative conditions. Moving well increases efficiency, which increases the number of repetitions that can be done before performance is compromised by fatigue or pain. Moving well also enhances performance by eliminating the weak links among all components of movement.

This book promotes the establishment of spine stability first, followed (sometimes) by spine mobility in some patients with back injuries. Although a small proportion of patients need mobilizing in a local spinal region first, most patients—from those looking for functional enhancement and pain relief to athletes seeking performance enhancement—benefit from stabilizing the spine first. In fact, once stability is established and the pain resolves, many patients find that their mobility returns with no further intervention (Moreside and McGill, 2012). Mobility elsewhere in the body linkage creates better back function. When testing back strength, remember that a good score requires a stable spine. In contrast, strong hips require good mobility. Poor hip function and control compromise back and torso function.

The evidence presented throughout this book is unanimous: A spine does not behave like a knee or shoulder, and approaches that work with these joints are often not effective for back therapy. Loading throughout the range of motion—which works well for joints in the extremities—is the nemesis of many backs, at least in the early stages of rehabilitation. During this period, training for strength is usually counterproductive. Unfortunately, the principles used in bodybuilding pervade rehabilitation clinical wisdom. This approach hypertrophies muscle at the expense of developing the functional motor and motion patterns needed for optimal health. This chapter provides

several general recommendations to maximize the chance for successful rehabilitation, followed by considerations of the stages of patient progression and guidelines for developing the best exercise regimen for each patient.

Pain-Reducing Mechanisms of Exercise

How does exercise reduce back pain? The mechanism of pain reduction is different for different types of back pain. When spine joints are unstable, abnormal micromovements irritate tissues. The stabilization exercises presented here have been shown to enhance stiffness (Lee and McGill, 2015), which reduces this aberrant motion. In addition, creating proximal stability enables controlled, high-quality movement distally, reducing tissue stress.

For specific movement intolerance, the irritation is reduced simply by avoiding the exacerbating movement. Those with identified tissue entrapment can reduce the pressure and pain with specific exercises. For example, we have shown that certain exercises can return displaced nucleus-impinging nerve roots (Scannell and McGill, 2009).

As previously noted, patients do not get used to pain, and continual tissue irritation increases their pain trigger sensitivity. Here the key is to desensitize the pain trigger, which is known as winding down the sensitivity. For those with fibromyalgic overlays to their back pain, in whom virtually everything causes pain, this means finding pain-free movements and loading patterns. Their virtual body map, including pain perception, has been distorted with trauma or disease (Butler and Mosely, 2013, provide a wonderful review). Repeating these exercise patterns teaches the brain pain-free movement. This process sometimes starts with left side–right side discrimination and imagined and reflective movement followed by actual body movement. Slowly, this pain-free movement repertoire expands.

Neurophysiological mechanisms of pain are highly modulated by movement and exercise. Endorphin release has long been known to occur with exercise, but the molecular basis for these links is not well understood. However, many more exercise-induced analgesic pathways are coming to light. One example is that exercise increases neurotrophin-3 production, which appears to have an analgesic role in cutaneous and deep tissue pain (Sharma et al., 2010).

Psychological components are important for some patients and influence the approach used to introduce and program exercise. For example, some patients obsess over their pain and what is going on in their bodies. Patients with high fear avoidance do better with physical therapy, which de-emphasizes anatomical findings (George et al., 2003). For these types of

patients, I suggest not using daily pain charts, but directing their focus to achieving good exercise form. Some patients have back pain because they overload themselves with daily regimens at the gym. For the type A personality, we ensure that exercise is not overexecuted (for people who believe that 25 repetitions of an exercise will do them more good than the 10 that were prescribed). For the type B personality who may bargain with the clinician about doing less exercise, performing the exercise dose as prescribed should be encouraged. These two personality types require different approaches to prescribing exercise.

Some pain is caused by what are referred to as sticky tissues. The concept that inflammation can cause sticky nerves was introduced for cases in which movement increased neural tissues' ability to slide with less pain-inducing friction. Fascial and muscular elements have been shown to get sticky and painful, and manual therapy approaches have been shown to help with this issue prior to movement exercise (Findley, 2011). Exercise has been shown to enhance pain-related ischemic conditions by promoting vascular sprouting (Wahl, Bloch, and Schmidt, 2007).

Some people have pain as a result of tissues that are inappropriately weak for the applied demands. In such cases the goal of exercise is to strengthen tissue, which requires a certain kind of programming. Athletes I have worked with who have recovered well from compression fractures have incorporated sufficient rest intervals to allow the injury to adapt between loading sessions. This appears to not work with collagenous disc injuries given the time required (years) for migrated nucleus material to gristle and form a plug (Adams and Dolan, 2005). Thus, the type of tissue that is the target for strengthening should determine the exercise approach.

Imbalances in joint and muscle function create odd stress patterns and more pain. For example, hip and back pain can cause gluteal muscle inhibition (Freeman, Mascia, and McGill, 2013), which makes the hamstring muscles dominant in creating hip extension moment. Because of the lines of action of these muscles, this causes more load on the anterior acetabulum and labrum of the hip (Lewis, Sahrman, and Moran, 2009), resulting in associated hip pain.

There is also substantial support for the notion of the existence of a psychological profile associated with fitness—that is, that fitter people ignore small discomforts. As discussed in the section *Why and How You Should Read This Book*, fitness reduces the tendency to catastrophize pain.

How is the most appropriate exercise determined to reduce each of these pain mechanisms? Although the assessment will reveal the pain mechanism, many challenging patients must be considered experiments in progress. Tweaking the exercise form and the dosage helps you to hone in on the optimal.

Five-Stage Back Training Program

We have developed a five-stage approach to back training that begins with identifying faulty movement patterns and ends with pain-free performance readiness. Consider this the big picture, because only athletes and people who perform demanding tasks will complete all five stages. However, it is important to understand the components and objectives of each stage, as well as their order. For example, patients may be unknowingly following a strengthening regimen (stage 4) without having addressed perturbed motion patterns (stage 1). Doing so will delay their recovery, or make them worse. So when approaching the program, you must first ask, Is the objective pain reduction and rehabilitation or athletic performance? Health objectives demand a focus on motion and motor patterns, stability, and endurance to achieve low tissue loads and a low-risk environment. Performance requires more overload, which naturally results in an elevated risk. The trick in the case of performance is to stay within the lowest risk possible.

Although all five major stages are listed here, only the first three are appropriate for rehabilitation and addressed in this book. Rigorous strength, speed, and power training are only for those interested in enhancing these attributes. The average person does not need them to have good back health. I address more advanced athletic training in detail in my book *Ultimate Back Fitness and Performance* (McGill, 2014). Begin with an assessment to understand the pain mechanisms. Then use the appropriate approach to address the mechanisms and build the person's tolerance and work capacity. Here is a summary of all five training stages:

- Stage 1: Establish quality movement engrams (motion patterns, motor patterns, and corrective exercise).
 - Identify perturbed patterns and develop appropriate corrective exercise.
 - Address basic movement patterns through to complex activity patterns.
 - Address basic balance challenges through to complex and specific balance challenges.
- Stage 2: Build whole-body and joint stability (focus on spine stability

here).

- Build stability while sparing the joints.
- Ensure sufficient stability commensurate with the demands of the task.
- Apply these patterns to the performance of daily activities.
- Stage 3: Increase endurance.
 - Address basic endurance training to ensure the capacity needed for stabilization.
 - Address activity-specific endurance (duration, intensity).
 - Increase the ability to repeat pristine movement patterns without fatigue-induced compromise.
 - Build the base for eventual performance training (only in those with this goal).
- Stage 4: Build strength.
 - Spare the joints while maximizing the neuromuscular compartment challenge.
 - Progress to skill movements.
- Stage 5: Develop speed, power, and agility.
 - Develop ultimate performance based on the foundation laid in stages 1 through 4.
 - Focus on optimizing elastic energy storage and recovery.
 - Employ the techniques of superstiffness.

If you master the first three stages, which are discussed in this chapter, you will understand how to get bad backs to respond and how to develop better rehabilitation programs.

Finding the Best Approach

Given the wide variety of patients with low back issues, we cannot expect to succeed in low back rehabilitation by treating everyone with the same cookbook program. The following strategies will help guide clinical decisions to individualize—and thus optimize—the rehabilitation program.

- Encourage patients to train for health versus performance. The notion that athletes are healthy is generally a myth, at least from a musculoskeletal point of view. Training for superior athletic performance demands substantial overload of the muscles and tissues of the joints. An elevated risk of injury is associated with athletic training and performance. Unfortunately, many patients observe the routines used by athletes to enhance performance and wrongly conclude that copying them will help their own backs. Training for health requires quite a different philosophy; it emphasizes muscle endurance, motor control perfection, and the maintenance of sufficient spine stability in all expected tasks. Although strength is not a goal, strength gains do result. If a patient with back pain states that his objective is to play tennis or golf, then he has the wrong short-term objective. First and foremost, the objective is to eliminate pain. Then the objective may shift toward a performance objective such as participation in a sporting activity.
- Integrate prevention and rehabilitation approaches. The best therapy rigorously followed will not produce results if the cause of the back troubles is not addressed. Part II provided guidelines for reducing the risk of back troubles: the importance of removing the cause of tissue overload cannot be overstressed. Linton and van Tulder (2001) demonstrated the efficacy of exercise for prevention; exercise satisfies the objective for both better prevention and better rehabilitation outcomes. First, teach patients what is causing their troubles; then work with them to eliminate the cause.
- Work toward a slow, continuous improvement in function and pain reduction. The return of function and the reduction of pain, particularly for those with chronic bad backs, is a slow process. The typical pattern of recovery is akin to that of the stock market. Daily, and even weekly, price fluctuations eventually result in higher prices. Patients have good

days and bad days. Many times lawyers have hired private investigators to make clandestine videos of people with back troubles performing tasks that appear inconsistent with those troubles. I am hired to provide comment. Some of these people are true malingerers and get caught. Others are simply having a good day when they are video-recorded. In such cases, I see all sorts of movement pathology consistent with their chronic history, and they are exonerated.

- Have the patient keep a journal of daily activities. Sometimes it is difficult to hone in on the pain mechanism and the correct dosage and exercise form. Examining daily pain and activity patterns can help identify the link with mechanical scenarios that exacerbate the pain. Two critical components should be recorded in a daily journal: how the back feels and what tasks and activities were performed. When patients encounter repeated setbacks, they should try to identify a common task or activity that preceded the pain episode. Likewise, even when progress is slow, patients should be encouraged to see some progress nonetheless. Without referring to the diary, patients sometimes do not realize that they are improving. Linking pain with a dose of activity is different from recording pain on a 10-point scale, which is typical of behavior modification programs. I have seen too many patients from these programs obsessing over their pain levels; for these people, we suggest stopping pain recording.
- Ensure a positive slope in progress. Chapters 10 and 11 introduce the big three exercises in different forms. We designed these exercises to spare the spine from large loads and to groove stabilizing motor patterns. Use the three to establish a positive slope in patient improvement. Once the slope is established, you may choose to add new exercises one at a time. The patient may tolerate some exercises well and others not so well. If the improvement slope is lost after adding a new activity, remove it, go back to the big three, and reestablish the positive slope. If the patient requires advanced exercises for athletic performance, perhaps to increase spine mobility, you may add exercises to achieve such objectives after establishing the positive slope. How long should each stage be? There is no single answer for everyone. Some progress quickly, whereas others require great patience. Your job is to determine the initial challenge, to gauge progress and enhance the challenge accordingly, and to keep the patient motivated, even during periods of

no apparent progress. The great clinicians blend keen clinical skills and experience with scientifically founded guidelines and knowledge.

- Determine whether the patient is willing to make a change. Obviously, the patient must change the current patterns that caused her to become a back patient. This requires motivation, which is not always easy to establish. Some have listed the importance of, and steps for developing, a change in motivation and attitude (e.g., Ranney, 1997). Briefly, such a program begins with the setting of goals—for example, returning to a specific job or partaking in a leisure activity. The employer's role in enhancing motivation is to ensure that modified work is available together with the opportunity for a graduated return to duty. Employers can also enhance motivation by fostering a culture in which worker success equates to company success, which in turn helps the worker. The second step in a motivation program is to formulate a realistic plan for reaching the goal established in the first step. It is beyond the mandate of this book to develop the components of maintaining and enhancing motivational opportunities at each stage of recovery.
- Determine whether the patient needs initial mobilization. Although everyone should incorporate spine stabilization exercises into daily activity, a small group of people will benefit from some directed soft tissue work (e.g., manipulation, trigger point therapy, Active Release Techniques, the use of foam rollers). These techniques are not the focus of this book. A word of caution is required here. Too many make the mistake of trying to mobilize a painful spine region that already has mobility. Nonetheless, there is good evidence that those with documented hypomobility may benefit from some initial manipulation or mobilization with a transition into stabilization training (Fritz, Whitman, and Childs, 2005).
- Consider other soft tissue treatments. A good manual medicine clinician may perceive local muscle spasms and odd-feeling local muscle texture. Further, these spasms and local neurocompartment disorders are associated with larger dysfunctions of the agonist and synergist muscles involved in a movement. In many cases these dysfunctions delay recovery or prevent complete recovery. Clinicians use a variety of soft tissue treatments to reduce spasm and release tissues that can impede attaining more normal muscular and joint function. Documenting them is beyond the scope of this book. We simply alert you to their potential

significance and role in rehabilitation.

- Avoid spine power. Spine power is the product of velocity and force ($\text{power} = \text{force} \times \text{velocity}$). This means that the spine is bending quickly and there is velocity in the muscles' lengthening and shortening. Techniques that involve high velocity in the spine have been shown to lead to back troubles, because they usually indicate high power (Marras et al., 1993; Stevenson et al., 2001). To minimize power and maximize safety, the forces transmitted through the trunk must be low if the spine is moving. If the forces transmitted through the trunk are high, then the velocity must be low. The power must be generated at the hips and shoulders and transmitted through an isometrically stabilized torso. Fortunately, this fundamental tenet for safety also helps to maximize performance.

Stages of Patient Progression

Before we can undertake to remove the activities that exacerbate low back troubles, we must determine what they are. This is a crucial part of the rehabilitation process. Uncovering the activities that cause back disorders begins with a patient interview. Some clinicians perform provocative testing at this time as well. Chapter 9 thoroughly discusses how to interview and test the patient. Once this has been done, rehabilitation can proceed. You may choose to overlap the stages involved in this process or conduct them in parallel. At all times, however, the objective is to establish and maintain the positive slope of continual improvement.

Most patients with chronic conditions do well with the universal principles of spine health: Move in a spine-sparing way throughout the day; develop appropriate spine and torso stiffness to eliminate painful spine micromovements; mobilize the hips; enhance endurance to maintain these pristine patterns throughout the day; know postures of comfort when you have overdone it; and know how and when to employ the movement techniques for a push, pull, lift, carry, squat, and lunge.

People in acute pain learn to settle their pain with pristine movements and by adopting appropriate pain-free postures (e.g., spine extension postures for posterior disc bulges). Finally, some conditions require special consideration, such as scoliosis and stenosis.

Stage 1: Establish Quality Movement Engrams

Some people are very body aware and can adopt a neutral spine or a flexed spine on command. Others can be frustratingly clueless. The objective of this stage is to detect and then correct perturbed movement patterns or engrams.

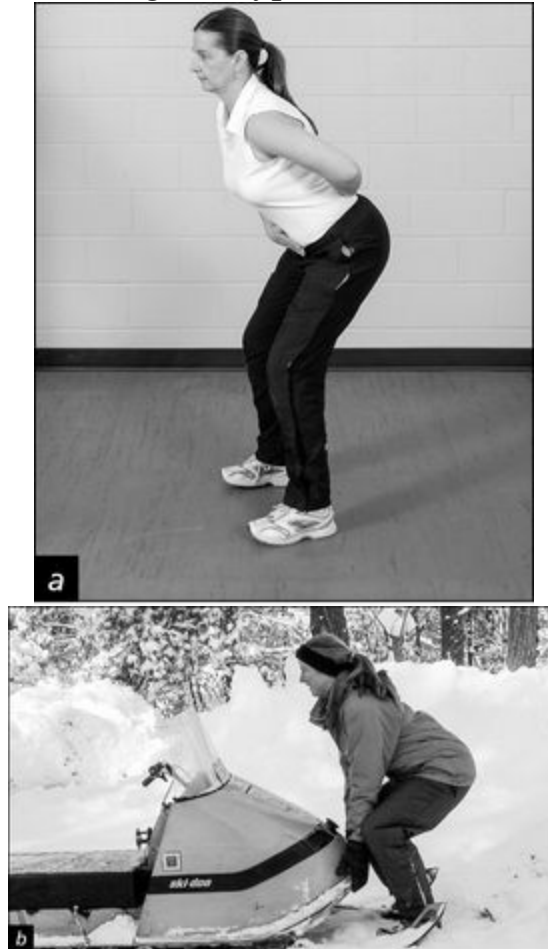
Distinguish Hip Flexion From Lumbar Flexion

The initial objective of stage 1 is to have the patient consciously separate hip rotation from lumbar motion when flexing the torso. For the more difficult cases, we typically begin by demonstrating on ourselves lumbar

flexion versus rotation about the hips. Other techniques that we have found particularly helpful are as follows:

- Have the patient place one hand on the abdomen while placing the other over the lumbar surface. This way the patient can feel whether the spine is locked and motion is occurring about the hips (see [figure 8.1a](#)).

Figure 8.1 For people who are not body aware and are unable to adopt a neutral or a flexed spine on command, we suggest (a) rehearsing the spine neutral position and hip (not lumbar) flexion while doing knee bends before (b) exertion, such as lifting this typical Canadian household item.



Photos from Stuart McGill

- Sometimes patients are receptive to being coached while using a practice load. The dummy constructed by nurses to help them rehearse proper patient lifting (shown in [figure 7.30](#) in chapter 7) is an example of such a practice load.
- Other patients respond well to photos of people correctly doing tasks that they will also be called on to do in the course of their jobs or their everyday activities. [Figure 8.2](#), a and b, shows examples of such photos.

Figure 8.2 Sometimes patients are receptive to task-specific illustrations showing spine and hip postures. (a) A rescue worker is applying a pulling force to the victim with the spine flexed. Discussion with the patient of this flexed spine posture together with one that would better spare the spine is very helpful. (b) Also discussed are exercise postures such as the neutral lumbar posture adopted by this patient performing cable pulls.



Photos from Stuart McGill

- Before (incorrect) and after (correct) photos, such as those in [figure 8.3](#), a through d, can be especially helpful.

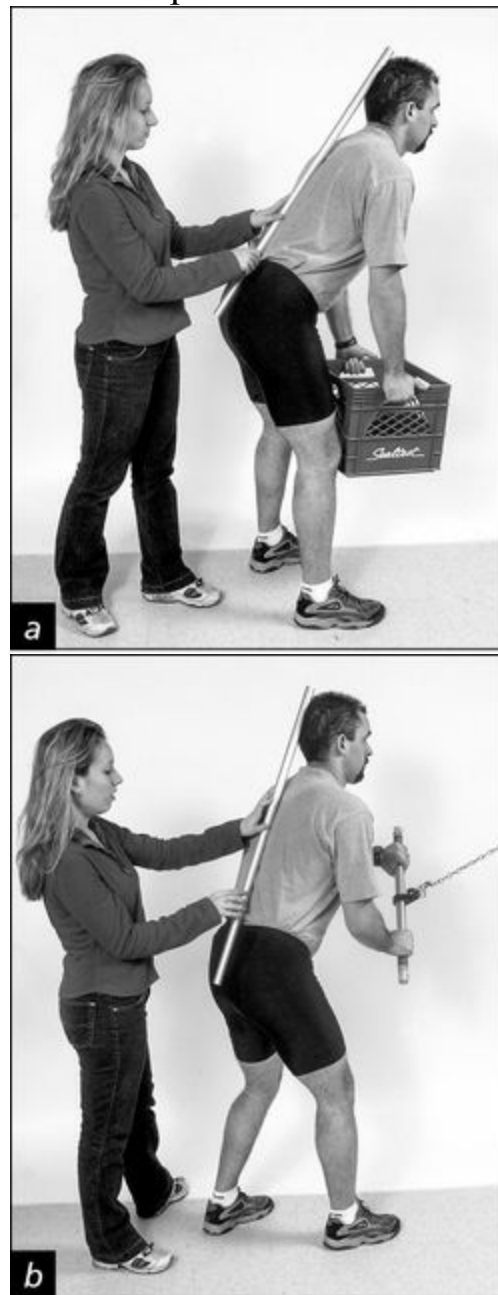
Figure 8.3 Workers relate to tasks with which they are familiar. These photos are helpful for construction workers, especially because they can readily see the difference between the (a, c) incorrect and (b, d) correct spine postures.



Photos from Stuart McGill

- Yet another technique is to place a stick along the spine with the instruction to flex the torso forward using the hips but maintain contact with the stick over the length of the spine (see [figure 8.4](#), a-c). Although this begins in the clinic, other, more complex bending tasks can also help groove these patterns into the general motion patterns used in work and other daily activities.

Figure 8.4 Placing a stick along the spine with the instruction to flex forward but maintain contact with the stick over the length of the spine helps patients separate hip rotation from lumbar motion when flexing the torso. Examples of motions that can be used are (a) sagittal plane symmetrical lifts, (b) three-dimensional cable pull exercises, and (c) work tasks that are familiar to the patient.

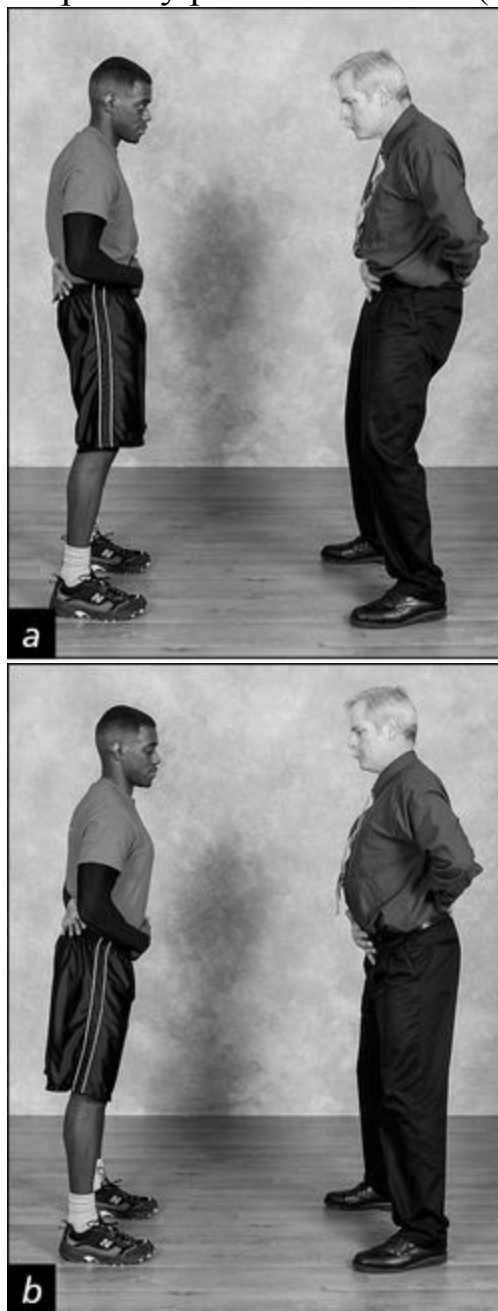




Photos from Stuart McGill

- When all of these attempts fail, we resort to the final technique: having the patient perform the “midnight movement” (rolling the pelvis)—this is lumbar motion. Interestingly, some patients who found sex painful never associated pelvis tilting with lumbar flexion (see [figure 8.5](#), a-d). Pointing this out to them often facilitates their making the next leap in spine position awareness and in being able to avoid these painful motions and postures.

Figure 8.5 Allowing the patient to feel spine (a) flexion and (b) extension helps them to eliminate these from the squatting motion. Others get it by (c) adopting the shortstop ready position and then (d) standing up.









Once you begin to see that any or several of the techniques just listed are helping these “difficult” patients learn how to achieve and maintain a neutral spine position during their daily activities, you may have them attempt some tasks to see whether the concept of the neutral spine is becoming ingrained. One such task is to take the spine stick and ask the patient to knock down an imaginary spiderweb in an overhead corner of the room. If he loses the lumbar neutral posture, point this out to him so he can correct it. Obviously, you should continue spine position awareness training with such a patient.

Some people have a very difficult time remembering the protective neutral spine pattern. We tell these types of patients to do the following:

1. Stop before an exertion (perhaps prior to lifting a household item).
2. Place the hands on the abdomen and lumbar region.
3. Practice a few knee bends with the motion about the hips and not the lumbar spine.
4. Then perform the lift.

This practice is effective for many people.

Three Movement Tools

For many people, learning to lock the rib cage on the pelvis is essential for injury prevention and for performance—though of course not for all. We have developed a teaching progression that is effective for most patients and is fully explained in chapter 8. However, we begin with three movement patterns that allow the patient to perform many daily activities without triggering pain (hip-hinge squat, lunge, and stop twist). Once these patterns are perfected, the patient learns when to use each tool (e.g., the lunge, not the squat, for moving to the floor).

- Hip-hinge squat. This pattern translates to rising from a chair or getting off the toilet, brushing teeth, and lifting. The motion is taken from the spine and transferred to the hips. The spine is stiffened into its pain-free posture (see [figure 8.6](#)).
- Lunge. This pattern assists in moving to and from the floor. No motion occurs in the painful spine (see [figure 8.7](#)).
- Stop twist. This pattern translates to people pushing and pulling on doors and performing other tasks in which twisting torque must be controlled so that the spine does not twist. Instead, the motion occurs at the ball and socket joints: the shoulders and hips (see [figure 8.8](#)).

The motion pattern should be accompanied by the abdominal brace motor pattern (see [figure 8.6](#)). More athletic versions of the squat progression are detailed in McGill (2014).

Figure 8.6 (a) The hip-hinge squat begins with placing the palms on the thighs. (b) The hands slide down the thighs and grab around the knees. Note that a vertical line through the knees falls between the heels and the balls of the feet. Now the spine is positioned into a neutral posture with cues such as "Lift your tail." (c) Then the torso weight is projected down the arms, which adds stiffness to the torso with latissimus and pectoral coactivation. (d) To cue the rise, the instruction is "Don't lift with the back, but pull the hips forward as you slide your hands up your thighs." This eliminates pain from bending, lifting, and so on. Test the person's competency by asking her to brush her teeth at the bathroom sink. Have her replicate the hip hinge and brush.



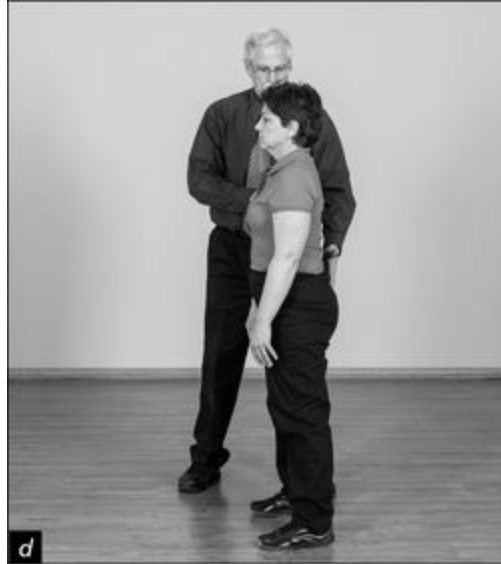
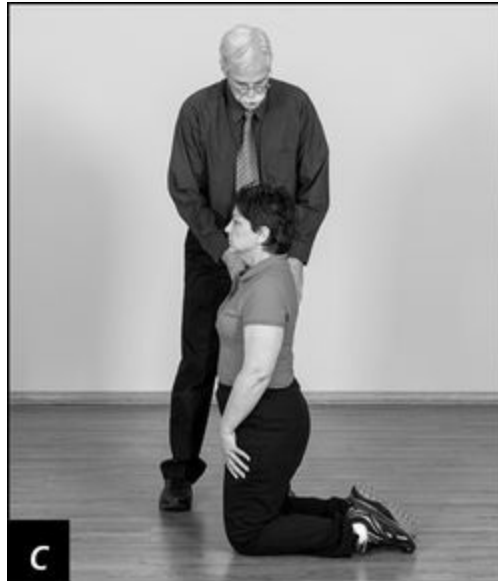


Figure 8.7 (a, b) The lunge begins by taking a step and descending until one knee is on the floor (note that no spine motion occurs at any stage during this movement). The torso is stiffened sufficiently to remove any pain-causing aberrant movement. (c) Then the person moves to both knees. The hands slide down the thighs by (d) hip hinging (no spine motion), and (e) the hands walk forward into a (f) hands-and-knees support posture. The person then rolls to the floor by sliding one hand forward and the same-side knee back.

The pelvis and rib cage drop to the floor in unison. (g, h) The roll is completed by lying on the back by stiffening the rib cage to the pelvis with an abdominal brace. (i, j, k) To rise, one leg remains straight while the other knee is raised together with the same-side arm. (l) As the roll continues, just as the knee is planted in the crawl posture, the shoulder is stiffened as the person rises on the support elbow. (m) The other knee is now pulled through with no spine motion. (n, o) The hip hinge is used to obtain an upright kneeling position from which the (p, q) lunge is used to get to the feet. The test is to ask the person to tie her shoe. From here she should lunge to a chair and take the hips to the target. Then she can tie her shoe with no spine motion or pain.











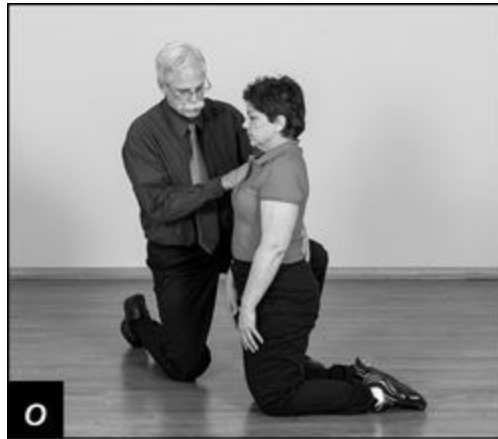




Figure 8.8 (a) The stop twist is a wall roll that begins with the person in the plank position with both elbows planted on the wall. (b) Many initiate the movement with the pelvis twisting and stressing the spine. (c) This is corrected when the focus is on spine and hip motion, generally adjusting the spine and hip posture to shift the pelvis toward the wall to find the sweet spot in spine position with minimal pain. (d , e) The abdominal muscles are braced and the rib cage is locked on the pelvis. The person pivots on the balls of the feet, pulling one elbow off the wall. No spine motion is permitted. These corrections are repeated until the person can control the spine and eliminate pain. The test is to then pull on a door, locking the rib cage to the pelvis; there should be no pain.







Distinguish Abdominal Bracing From Abdominal Hollowing

Maintaining a mild contraction of the abdominal wall can also help ensure sufficient spine stability. This maneuver is called abdominal hollowing in many back care circles; however, we prefer to avoid that terminology, because it suggests to most people either pulling in or puffing out the abdominal wall. When the contraction is performed correctly, no geometric change at all occurs in the abdominal wall. In other words, rather than hollowing or drawing in the abdominal wall, the patient simply activates the muscles to make them firm and stiff. We call this contraction abdominal bracing.

There is confusion in the interpretation of the literature regarding abdominal hollowing and abdominal bracing is discussed in chapter 5. Abdominal bracing, which activates the three layers of the abdominal wall (external oblique, internal oblique, transverse abdominis) with no drawing in, is much more effective than abdominal hollowing at enhancing spine stability

(Brown and McGill, 2008, 2010; Brown, Vera-Garcia, and McGill, 2007; Grenier and McGill, 2007; McGill, 2001) (see [figure 8.9](#)). An important issue is the intensity. Bracing effort runs the full continuum, from very mild for those intolerant of compressive load but need some control (roughly 2 to 5% of maximal contraction), to lifting and carrying a load that must be controlled (5 to 20% of maximal contraction), to those maximizing storage and recovery of elastic energy (about 25% of maximal contraction), to those performing a maximal-effort lift (well over 50% of maximal contraction).